Quantum Fault Tolerance Meets Toom's Contours

Michael Ben-Or David Ponarovsky

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1 Errors Clusters

[COMMENT] We should mention, that in any time that we call to $H^{\otimes n}$, even if its only for moving to the dual space, we immediately apply after the ϕ-transpose mapping. The idea is that it will allow to us to ensure that the X-errors are always bounded by upper-right tingle and the Z-errors are always bounded by lower-left triangles.

Definition 1.1 (**Size**()-Function)**.** Consider a (d, d')-Toric complex $\mathcal{G}$, and let S be a subset of vertices, $S \subset \mathcal{G}$. We define the *size* of S to be:

$$\mathbf{Size}(S) := \sum_{i}^{d+1} \max_{v \in S} L_i(v)$$

Furthermore, we extend the notion to binary assignment over the ith faces of a complex $z : \mathcal{F}^{(i)} \rightarrow \mathbb{F}_2$ by associating $\mathbf{Size}(z)$ with the size of the vertex subset supporting z.

Claim 1.2. Let $\mathbf{S} = \{S_1, S_2, .., S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^{k} S_i) \leq \sum_{i=1}^{k} \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned}
\mathbf{Size}(R_1 \cup R_2) &= \sum_{i}^{d+1} \max a_{i,1}, a_{i,2} = \sum_{i}^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\
&\leq \sum_{i}^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\
&= \mathbf{Size}(R_1) + \mathbf{Size}(R_2)
\end{aligned}$$

$\square$

Definition 1.3. We say that a Pauli operator P is an X*-type error* if it consists of a multiplication of single-qubit Pauli X and identities. Similarly, we say that P is a Z*-type error* if it consists of a multiplication of single-qubit Pauli Z and identities. For a binary assignment over the d'-faces, $z : \mathcal{F}^{(i)} \to \mathbb{F}_2$ we say that z is the *underline assignment* of P if z is the support of the non-trivial coordinates of P. We extend the size function to X/Z-types errors as follow:

1. If P is a X-type error, then $\mathbf{Size}(P) := \mathbf{Size}(z)$.
2. Else, namely P is a Z-type error, then $\mathbf{Size}(P) := \mathbf{Size}(\phi(z))$.

Claim 1.4. The following hold:

1. $\phi \circ H^{\otimes n}$ sends X-type triangles to the same Size Z-type triangles, and vice versa.
2. $\mathbf{CX}$ and φ send X-type triangles to the same size X-type triangles. Similarly, they send Z-types triangles to the same size Z-type triangles.

Proof. Let z be a binary assignment over the d'-faces, $z : \mathcal{F}^{(i)} \to \mathbb{F}_2$, and let P be the X-type error which z is its underline assignment. Let $Q = \phi \circ H^{\otimes n} P$, and denote by z_Q the underline assignment of Q.

$$\begin{aligned} Q\left|\psi\right\rangle &= \phi \circ H^{\otimes n} P \left|\psi\right\rangle = \phi \circ H^{\otimes n} X_z \left|\psi\right\rangle \\ &= Z_{\phi(z)} \phi \circ H^{\otimes n} \left|\psi\right\rangle = Z_{\phi(z)} \left|\psi_2\right\rangle \end{aligned}$$

Where $\left|\psi_2\right\rangle$ is a codeword. Thus, Q is a Z-type error with underline assignment $z_Q = \phi(z)$, thus:

$$\mathbf{Size}(Q) = \mathbf{Size}(\phi(z_Q)) = \mathbf{Size}(\phi^2(z)) = \mathbf{Size}(z) = \mathbf{Size}(P)$$

For the second part, we consider two registers, R_1 and R_2. The $\mathbf{CX}$ acts transversally on the registers, and it's clear, by commutation relations, that it duplicates X-type errors from R_1 to their counterparts on R_2. Similarly, $\mathbf{CX}$ duplicates Z-type errors from R_2 into their counterparts on R_1.

$$\mathbf{Size}(\varphi P) = \mathbf{Size}(\varphi(X_z)) = \mathbf{Size}(X_{\varphi(z)})$$

□

Claim 1.5. Let R_1, R_2 be two registers, then any logical operator in $\mathcal{O}$ sends error cluster to an error cluster as follows:

2 Toric Encoded Circuits

Definition 2.1 (Troic Encoded Circuit)**.** Let C be a quantum circuit. We say that C is *Toric encoded* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \ldots, R_k$, and each of the registers is encoded by $\mathcal{C}$.
2. The gates in C with even time coordinates are sweep decoding gates.

For a set of faults $\mathcal{E}$ the gates in $C[\mathcal{E}]$ can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, .., D_1 P_1 \Lambda_1$$

Definition 2.2 (Base graph G_o)**.** Let C be a quantum circuit, at depth d, with registers $R_1, \ldots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, .., \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C, denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigsqcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$\begin{aligned} E_o := \Big\{ \{(v,t),(u,t')\} : \text{ if} \\ &\text{either there exists } i \text{ such } \{v,u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ &\text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1}+1 \Big\} \end{aligned}$$

[COMMENT] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 2.3 (Time Graph of $C(\mathcal{E})$)**.** For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o. For each time t, denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial \mathcal{D}_t$ is the syndrome that the qubits in the deviation admit. We define the history graph $\mathcal{H} = (V_\mathcal{H}, E_\mathcal{H})$ as follows:

1. The vertices $V_\mathcal{H}$ are taken to be the vertices of G_o, namely tuples $(v,t) \in V_o$, that satisfy that v belongs to a face associated with a check, which, at time t, belongs to the $\partial \mathcal{D}_t$. Namely:

$$\Big\{ v_t : v_t = (v,t) \in V_o \text{ and there exists a check } f \in C \text{ such } v \in f \in \partial \mathcal{D}_t \Big\}$$

2. The edges $E_\mathcal{H}$ is partitioned into two types $A_\mathcal{H}$ and $F_\mathcal{H}$, where $A_\mathcal{H}$ are the edges in E_o restricted to $V_\mathcal{H}$ and $F_\mathcal{H}$ are defined to be the . Namely:

$$\begin{aligned} A_\mathcal{H} &= \Big\{ e = \{u,v\} \in E_o : u, v \in V_\mathcal{H} \text{ and } T(v) = T(u) \pm 1 \Big\} \\ F_\mathcal{H} &= \Big\{ e = \{u,v\} \in E_o : \text{ exists } w \in V_\mathcal{H} \text{ such } \{v,w\}, \{u,w\} \in A_\mathcal{H} \Big\} \end{aligned}$$

By definition of the construction, $G_{\mathcal{H}}$ is a time-graph.

Definition 2.4 (Size Invariant Action)**.** [COMMENT] Here we classify both transversal gates, and rotation on the Toric. An action g is be said size invariant action if, for any assignment z, it holds that $\mathbf{Size}(gz) = \mathbf{Size}(z)$.

Claim 2.5. The sweep rule is Size ξ-shrinking.

Proof. $\square$

[COMMENT] Eventually, we might not used the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time $= 3t$, in that we apply an invariant action so.